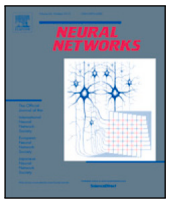
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Full Length Article



# Self-supervised pre-trained neural network for quantum natural language processing

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ABSTRACT

Quantum computing models have propelled advances in many application domains. However, in the field of natural language processing (NLP), quantum computing models are limited in representation capacity due to the high linearity of the underlying quantum computing architecture. This work attempts to address this limitation by leveraging the concept of self-supervised pre-training, a paradigm that has been propelling the rocketing development of NLP, to increase the power of quantum NLP models on the representation level. Specifically, we present a self-supervised pre-training approach to train quantum encodings of sentences, and fine-tune quantum circuits for downstream tasks on its basis. Experiments show that pre-trained mechanism brings remarkable improvement over end-to-end pure quantum models, yielding meaningful prediction results on a variety of downstream text classification datasets.

## 1. Introduction

Recent years have witnessed a booming rise of quantum computing, with rapid development on both quantum hardware and quantum algorithms. The intrinsic computational speed-ups in the fundamental concepts of superposition and entanglement have led to powerful applications in a wide range of areas, such as chemistry (Cao et al., 2019; McArdle, Endo, Aspuru-Guzik, Benjamin, & Yuan, 2020), image processing (Lloyd, Schuld, Ijaz, Izaac, & Killoran, 2020; Mari, Bromley, Izaac, Schuld, & Killoran, 2020; Mogalapalli, Abburi, Nithya, & Bandreddi, 2021; Oh, Choi, & Kim, 2020), cryptography (Mavroeidis, Vishi, Mateusz, & Jøsang, 2018; Proos & Zalka, 2004) and machine learning (Jaderberg et al., 2022; Li, Wang, Zhu, Lioma, & Liu, 2023; Li, Zhao, & Wang, 2024). Despite the progress, quantum devices are still living in the Noisy Intermediate-Scale Quantum (NISQ) Era (Preskill, 2018). Due to the inherent noises in the hardware, current devices can support practical implementations of quantum algorithms with a scale of up to 50–100 qubits.

Therefore, researchers are interested in developing practical quantum algorithms for NISQ devices. Inspired by the power of machine learning, quantum machine learning (QML) (Biamonte et al., 2017) has emerged as a popular research topic. Notably, parameterized quantum circuits (PQCs) are constructed for machine learning tasks, encoding input features to quantum states and making predictions with measurements. Various algorithms have been devised to train the quantum circuits from the data in a supervised manner (Liao, Hsieh, & Ferrie, 2021).

As a key subfield of AI, natural language processing (NLP), the field that aims at analyzing and understanding human language, is arguably one field that so far has been poorly explored in QML (Di Sipio, Huang, Chen, Mangini, & Worring, 2021). To date, current pure Quantum NLP (QNLP) models (Coecke, de Felice, Meichanetzidis, & Toumi, 2020; Kartsaklis et al., 2021a; Lloyd et al., 2020; Lorenz, Pearson, Meichanetzidis, Kartsaklis, & Coecke, 2021; Meichanetzidis et al., 2020; Zeng & Coecke, 2016) are mainly based on Categorical Compositional Distributional (DisCoCat) model (Coecke et al., 2020), a mathematical framework that encodes syntactical structure of sentences in the form of tensor network, where individual word representations are aggregated to sentence representations along the structure in a bottom-up fashion (Lorenz et al., 2021). The tensor networks can be efficiently converted to PQCs that support end-to-end supervised learning in real datasets (Lorenz et al., 2021).

The DisCoCat-family models are limited in two aspects despite their preliminary success on toy text classification tasks (Lorenz et al., 2021). On the one hand, it scales poorly, since it requires a syntactical parsing of sentences, which is prohibitive and error-prone when the sentence length gets large. On the other hand, as a pure quantum computing model, it suffers from a limited capacity due to the high linearity of quantum gates and circuits (Schuld & Petruccione, 2021). One

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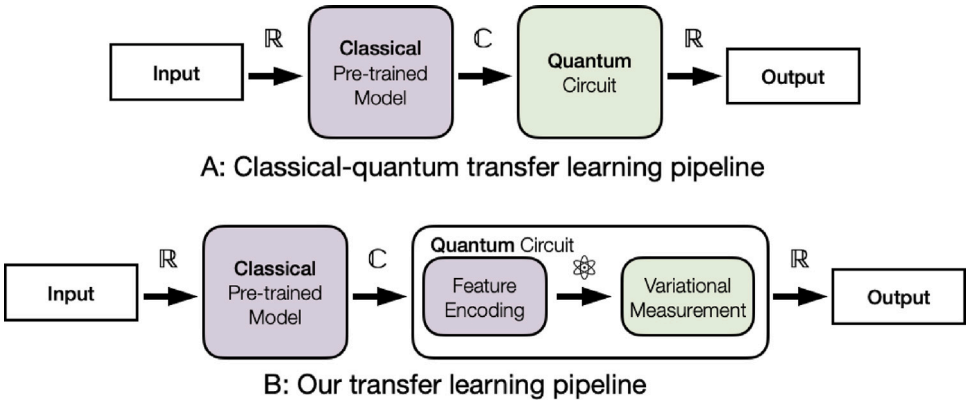



**Fig. 1.** A: Classical-quantum transfer learning pipeline (Mari et al., 2020). A classical pre-trained model (purple box) is act as a fixed feature extractor, followed by a not pre-trained quantum circuit (green box) which is optimized for the specific task. B: Our transfer learning pipeline. In comparison to the classical-quantum transfer learning pipeline, we add a pre-trained and trainable feature encoding quantum module to better fit downstream tasks.

attempt to address the limitations of DisCoCat-family models is hybrid classical-quantum training (Li et al., 2024), which entails interleaving classical and quantum layers in the architecture, and introduces non-linearity in classical layers. By design, it achieves remarkably enhanced effectiveness over DisCoCat models. However, the frequent switching between classical and quantum layers lowers the speed of training and inference, severely undermining the advantage of quantum computing architectures in practical applications.

We posit that a more promising direction is the exploration of pre-training strategies to obtain powerful quantum language representations. Self-supervised training has been the core mechanism to the development of pre-trained language models (PLMs) (Brown et al., 2020; Devlin, Chang, Lee, & Toutanova, 2019) and more recently large language models (LLMs) (Achiam et al., 2023; Wu et al., 2023), both achieving milestone success on human language understanding. Inspired by this success, we are motivated to explore *whether self-supervised pre-train mechanism can benefit QNLP models by providing powerful quantum encodings of language*.

Related works (Li et al., 2023; Qi, Yang, Chen, & Hsieh, 2023; Yang, Qi, Chen, Tsao, & Chen, 2022) have provided indirect evidences supporting this claim: Qi et al. (2023) put forth a pre-training mechanism for tensor-train network-variational quantum circuit (TTN-VQC) that effectively improves the representation and generalization power of VQC, Yang et al. (2022) attain strong performance while enhancing data protection on spoken language intent classification tasks by building a quantum temporal convolution (QTC) model on top of the classical BERT embedding. However, the research question can only be best answered by proposing a dedicated self-supervised mechanism for pre-training quantum representations of language, which still remains unexplored. Li et al. (2023) pre-train a complex-valued BERT with a unitary classification (i.e. CLS) head, achieving comparable fine-tuning performance to original BERT on GLUE benchmarks. Despite its claim that the model is quantum computing compatible, however, its classification head cannot be implemented with a quantum circuit, since the unitary matrix in the classification head cannot be efficiently converted to a hardware-efficient *ansatz*. The observed accuracy values do not indicate the authentic capacity of a pure quantum NLP model under pre-trained quantum encodings.

In this work, we proposed a concise yet effective model framework, which benefits from the classical-quantum transfer learning method (Fig. 1). We pre-trained our models on extensive text corpora leveraging self-supervised learning, aiming to assist the model in acquiring linguistic knowledge. The pre-trained model exhibits strong performance across diverse classification tasks necessitating varying linguistic competencies. The source code for this work is publicly available on github [1].

[1] https://github.com/qiuchili/qbert

**We contribute** the first approach to introduce the classical-quantum transfer learning for QNLP models, and pre-train language representations compatible with quantum computing models. Apart from the remarkably improved performance, our model is more scalable than existing QNLP models and can tackle longer sentences.

## 2. Related work

In this work, we investigate a self-supervised pre-training paradigm for quantum NLP models. On the one hand, our work is inspired by the success of existing self-supervised models and pre-training method, which both flourish in classical and quantum fields. On the other hand, our work is also a supplement to the current QNLP research, in which there are still many unexplored facets. This section overviews the past and present research within the QNLP field, and broadly surveys the related works on self-supervised learning and pre-training.

### 2.1. Quantum natural language processing model

As an important branch of QML, QNLP has gradually evolved into different strands of research. Broadly speaking, QNLP models can be categorized into two types: (I) pure QNLP models and (II) hybrid quantum–classical models. The DisCoCat model for natural language, which relies on a syntax-aware algorithm that incorporates the information regarding compositional language structure and meaning of sentences (or phrases) into the construction of quantum networks (Guarasci, De Pietro, & Esposito, 2022), serves as the shared theoretical framework across almost all pure QNLP approaches (Coecke, Sadrzadeh, & Clark, 2010; Guarasci et al., 2022). Based on pregroup grammar (Lambek, 1958, 1999), models under the DisCoCat framework transform sentences into string diagrams, which serve as a network-like language (Coecke & Kissinger, 2018). These diagrams are then converted to quantum circuits, and the trainable parameters within the model. In the context of small-scale QNLP tasks, the pure QNLP models have shown promising results by addressing numerous issues associated with well-established NLP mechanisms (Coecke et al., 2020; Guarasci et al., 2022; Kartsaklis et al., 2021a; Lloyd et al., 2020; Lorenz et al., 2021; Meichanetzidis et al., 2020; Zeng & Coecke, 2016).

In addition to the pure QNLP models, hybrid quantum–classical models, which are a combination leveraging the advantages of both classical machine learning algorithms and quantum computing (Pandey, Basisth, Sachan, Kumari, & Pakray, 2023), represent another promising research direction in the field of QNLP. Benefiting from the included classical network components, hybrid QNLP models can handle relatively large-scale textual data by constructing appropriately, and have exhibited strong performance on some datasets (Di Sipio, Huang, Chen, Mangini, & Worring, 2022; Li et al., 2024; Yang et al., 2022).

### 2.2. Self-supervised pre-training model

The increase in data volume leads to an expensive cost of manual annotation, posing a fundamental challenge for supervised learning. Furthermore, supervised learning faces issues like generalization error and spurious correlations (Liu et al., 2021). Therefore, researchers are turning to seek alternatives, and self-supervised learning emerges as a solution without any prohibitive labor costs (Rani, Nabi, Kumar, Mittal, & Kumar, 2023). Self-supervised learning is a paradigm for training models to learn valuable and knowledgeable information from large-scale unlabeled data. It relies on the implicit properties of data to introduce the self-defined labels, which work as supervisions for the follow-up training, eliminating the need for any manual annotations (He, Deng, Zheng, Li, & Situ, 2023). It has been applied in numerous models and has achieved impressive accomplishments in both the computer vision (CV) and NLP fields (Chen, Kornblith, Norouzi, & Hinton, 2020; Devlin et al., 2019; He, Fan, Wu, Xie, & Girshick, 2020). In NLP-related research, self-supervised learning is extensively employed for pre-training language models, aiming to extract implicit syntactic, grammatical, semantic, and other linguistic understanding from large corpora in a more cost-effective manner. Conneau et al. (2019), Devlin et al. (2019), Liu et al. (2019), Radford and Sutskever (2018), Yang et al. (2019).

In the field of quantum machine learning, self-supervised learning has led to significant advances, especially in the quantum image processing domain. Various self-supervised quantum learning networks with diverse architectures have been applied in the field of medical imaging analysis, particularly in the diagnosis of brain tumors, yielding promising results (Konar, Bhattacharyya, Dey, & Panigrahi, 2019; Konar, Bhattacharyya, Gandhi, & Panigrahi, 2020; Konar, Bhattacharyya, Gandhi, Panigrahi, & Jiang, 2023; Konar, Bhattacharyya, Panigrahi, & Behrman, 2021; Suneel et al., 2024). Notably, the hybrid classical-quantum visual model trained in a self-supervised manner performs a numerical superiority over the classical neural network when both the sizes of the quantum neural network and the equivalently structured classical network are limited (Jaderberg et al., 2022). However, self-supervised pre-training has yet to be explored for quantum natural language processing to our best knowledge. Li et al. (2023) claim to pre-train a quantum-compatible complex-valued BERT with a unitary classification (i.e. CLS) head. However, the classification head cannot be implemented with a quantum circuit, so it cannot be executed on a quantum device in either pre-training or fine-tuning.

## 3. Background

In this section, we introduce the background of quantum computation, which is necessary for understanding our model and method. We will elaborate on the basic definitions of qubits, quantum gates, and quantum circuits.

### 3.1. Qubit

#### 3.1.1. Single qubit

The fundamental unit of quantum computation is *quantum bit*, or *qubit* in Hilbert space. Unlike classical bits that can only be in state 0 or 1, a qubit can exist in a superposition of 0 and 1, represented as:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,. \tag{1}$$

The $|\psi\rangle$ can be called a *pure state* of a qubit. The parameters $\alpha$ and $\beta$ are both complex numbers, called *amplitudes* of the *basis* states $|0\rangle$ and $|1\rangle$. They meet the normalization constraint $|\alpha|^2 + |\beta|^2 = 1$.

Usually, we cannot tell the exact quantum state of a qubit directly. Nevertheless, we can obtain restricted information about the quantum state of a qubit, that is, the probability of the qubit *collapsing* to either 0 or 1. When we *measure* a qubit, it collapses onto either $|0\rangle$ or $|1\rangle$. When the measurement is repeated for sufficiently many times, the measurement result will be 0 for $|\alpha|^2$ of the measurements and 1 for $|\beta|^2$ of the measurements.

#### 3.1.2. Multiple qubits

In a qubit system, there can also be multiple qubits. Taking a two-qubits system as an example. The possible basis states of this system include $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$. The possibility of the two-qubit system being in any of the mentioned four states is valid. Therefore, similar to a single qubit system, this two-qubit system can be represented by the following equation:

$$|\psi\rangle = \alpha_{00}\,|00\rangle + \alpha_{01}\,|01\rangle + \alpha_{10}\,|10\rangle + \alpha_{11}\,|11\rangle\,. \tag{2}$$

The complex number $\alpha_{ij}$ refers to the amplitude of the corresponding quantum state, which meets the normalization constraint $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$. With the gradual increase of the number of qubits in a quantum system, the number of possible quantum states also exhibits exponential growth. For a system with $n$ qubits, it is located in a $2^n$-dimensional Hilbert space, indicating the existence of $2^n$ possible quantum states.

### 3.2. Quantum entanglement

*Quantum entanglement* describes a special correlation between two or more qubits, where the quantum states of the qubits cannot be described independently, even when they are spatially separated. For two entangled qubits, an operation on one qubit will result in a corresponding change in the state of the other qubit (Liu, Li, Wang, Zhang, & Song, 2023).

Mathematically, if a system cannot be decomposed into a tensor product of its local constituent states, it is defined as an entangled system; in other words, they are not individual qubits but rather an inseparable whole. Given two arbitrary quantum systems $A$ and $B$ with their respective Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, the Hilbert space of the composite system $|\psi_{AB}\rangle$ is the tensor product of these spaces:

$$|\psi_{AB}\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B. \tag{3}$$

If $|\psi_{AB}\rangle$ is an entangled state, it cannot be decomposed into the tensor product between state $|\psi_A\rangle$ and $|\psi_B\rangle$:

$$|\psi_{AB}\rangle \neq |\psi_A\rangle \otimes |\psi_B\rangle\,. \tag{4}$$

### 3.3. Quantum gates

To manipulate the quantum state of a qubit system, some operations, such as rotations, flips, exchanges, etc., are applied over the qubits in the system. More specifically, these implementations depend on the *quantum gates* practically. If we describe a quantum gate mathematically, it can be considered as a *unitary* matrix consisting of complex numbers. The *unitarity* constraint signifies that for a quantum gate $U$, the equation $U^\dagger U = I$ must hold, where $U^\dagger$ represents the conjugate transpose of $U$, and $I$ is the identity matrix. In quantum computation, these quantum gates, whether complex or simple, are essential cornerstones for manipulating quantum information.

In our project, we mainly use four quantum gates. The single-qubit *rotation* gates about the $x$, $y$ and $z$ axes, which are utilized to rotate the qubits:

$$R_x(\theta) \equiv \begin{bmatrix} cos\frac{\theta}{2} & -i\,sin\frac{\theta}{2} \\ -i\,sin\frac{\theta}{2} & cos\frac{\theta}{2} \end{bmatrix} \quad R_y(\theta) \equiv \begin{bmatrix} cos\frac{\theta}{2} & -sin\frac{\theta}{2} \\ sin\frac{\theta}{2} & cos\frac{\theta}{2} \end{bmatrix}$$

$$R_z(\theta) \equiv \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

And the two-qubit *controlled NOT* ($CNOT$) gate which helps us to entangle qubits:

$$U_{CNOT} \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

In quantum computation, according to the *Z-Y Decomposition Theorem* for a single qubit, an arbitrary quantum gate $U$ for one single qubit can be decomposed in to a specific combination of two $R_z(\theta)$ gates and one $R_y(\theta)$ gate (Nielsen & Chuang, 2010):

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta). \tag{5}$$

In Eq. (5), $\alpha$, $\beta$, $\gamma$ and $\delta$ are all real numbers and the $e^{i\alpha}$ is the *global phase shift*. In our research, the global phase shift $e^{i\alpha}$ is ignored, as its transformation does not alter the probability distribution of measurement results. We leverage this theorem to manipulate a single qubit within a quantum circuit using $R_z$ and $R_y$ gates.

### 3.4. Quantum circuit

The quantum circuit, comprised of qubits and quantum gates, is utilized to describe algorithms and operations in quantum computing. The wires in the quantum circuit are excited and entangled through a series of quantum operations, and they are measured to determine the probability of all possible quantum states. Measurement causes the collapse of superposition into a definite state, projecting a qubit $|\psi\rangle$ into either $|0\rangle$ or $|1\rangle$. For an $n$ qubit system, if we perform enough measurements, we will obtain a $2^n$-dimensional probability distribution vector, where each position corresponds to the probability of a possible quantum state.

Although theoretically, we can construct quantum circuits with arbitrary structures and numbers of qubits, we often resort to using hardware-efficient circuits due to the limitations of quantum hardware in reality. In our project, we employ two pre-built quantum circuit layouts which consist of 1-qubit gates and 2-qubit gates, to increase the compatibility of our model with quantum hardware. These layouts of quantum circuits are also referred to as *ansatz*. By adjusting the parameters within these parameterized quantum architectures, a series of possible quantum states can be obtained.

## 4. Model

In this section, the architecture of Quantum Compatible BERT (QCBERT) model and its implementation for both pre-training and fine-tuning phases are introduced. In order to learn meaningful text representation amenable to quantum computing, QCBERT is designed to be a hybrid classical-quantum model and is jointly pre-trained by both self-supervised tasks: Next Sentence Prediction (NSP) and Masked Language Model (MLM). Particularly, the NSP classification head is designed as a parameterized quantum circuit, encoding the whole text sequence as an authentic quantum state and predicting the NSP label with quantum measurement. During fine-tuning, the parameters of the multi-layer Transformer encoder and feature extractor are inherited from the associated modules in the pre-training phase, while the task-related parameters in the classifier are randomly initialized and trained in a data-driven manner. The pipeline of pre-training and fine-tuning is illustrated in Fig. 2, while the main steps therein are further described in the following text.

### 4.1. Model architecture

The QCBERT model contains three modules: a multi-layer Transformer encoder, a feature extractor and a task-related classifier, as shown in Fig. 2. In the pipeline, an input sentence is encoded to a fixed-dimensional vector by the QCBERT encoder, which is further converted to a multi-qubit quantum state by the feature extractor. The state then works as the input to the variational measurement circuit in the task-related classifier, where the Variational Measurement Ansatz produces a probability distribution of all basis states as the measurement result. The probability vector is finally passed to a linear projection layer with trainable parameters to obtain the predictions.

#### 4.1.1. QCBERT encoder

The encoder maps any input sentence to a fixed-dimensional vector. Even though it can be any neural network, we adopt the BERT encoder with a multi-layer Transformer architecture to fully exploit the power of language model pre-training. Essentially, it contains an embedding layer with a maximum sequence length of 512 followed by 12 Transformer layers of 12 attention heads each. The hidden size of Transformer blocks is set to be 768 and the intermediate size in the feed-forward network is set to be 3072. In pre-training, the parameters in the QCBERT encoder are initialized from an existing BERT-base checkpoint and continuously pre-trained with the pre-training corpora.

#### 4.1.2. Feature extractor

The feature extractor aims at constructing the quantum sentence encoding from the encoded token representations, which is an open question. Crucially, the state should not only be a unit-length complex vector in its mathematical form, but also support efficient implementation via quantum hardware. Therefore, a quantum circuit is implemented with a hardware-efficient ansatz $U_\Theta(x)$ composed of $l_{FE}$ deep basic entangler layers with an input state of $|0\rangle$, i.e. $|S\rangle = U_\Theta(x)|0\rangle$. In contrast, Li et al. (2023) directly learn the unit-length complex vector $|S\rangle$ in pre-training, but it cannot be efficiently mapped to a quantum circuit. Hence it is not a quantum computing compatible pre-training mechanism as claimed.

As shown in Fig. 3 A, each layer comprises single-qubit rotations about the $x$ axis which only include one parameter per qubit, and *CNOT* gates which entangle each qubit with its neighbor.[2] This architecture repeats $l_{FE}$ times in a cascaded manner in the circuit to strengthen the entanglement among qubits. The dimension of the parameter space of the Feature Encoding Ansatz depends on the number of qubits $n$ and the depth of the quantum circuit $l_{FE}$. Here we opt for $n = 5$ and $l_{FE} = 10$, resulting in a parameter space dimension of (10,5). We will discuss the different settings of the parameters in detail in Section 5.4.

We seek to project the hidden states obtained from the QCBERT encoder to the parameter space of Feature Encoding Ansatz. In light that the [CLS] token is widely regarded and employed as the sentence representation, the hidden state of [CLS] token is fed into a fully-connected layer to match the dimension of parameter space in the Feature Encoding Ansatz. These newly obtained values are rescaled to range $[0, 2\pi)$ and are then reshaped to (10,5) to finally produce the angles $\Theta = \{\theta_{1:n}^{1:l_{FE}}\}$ for rotations about the $x$ axis. Each $\theta$ value corresponds to a unique qubit id at its subscript and a layer id at its superscript. The complete parameter acquisition process can be represented by the following formula, where $f(\cdot)$ denotes the mapping by the linear projection layer and $\sigma(\cdot)$ refers to the sigmoid function:

$$\Theta = 2\pi \cdot \sigma(f(\mathbf{h}_{[\mathrm{CLS}]})). \tag{6}$$

#### 4.1.3. Task-related classifier

The architecture of the Variational Measurement Ansatz in classifier is constructed from single-qubit gates that locally manipulate each individual qubit and two-qubit gates that introduce entanglement between different qubits. this quantum circuit $U_\Phi(x)$ is implemented with $l_{VM}$ strongly entangling layers (Schuld, Bocharov, Svore, & Wiebe, 2020). As shown in Fig. 3 B, each layer carries out single-qubit rotations about the $z$, $y$ and $z$ axes, and then entangles all $n$ qubits by using ($n$) sequentially arranged *CNOT* gates. The $i$th ($i$ is an integer and $1 \le i \le n$) qubit in the $j$th ($j$ is an integer and $1 \le j$) strongly entangling layer is connected with the $((i+j) \mod n)$th qubit by *CNOT* gate.[3] The input to the Variational Measurement Ansatz is the encoded state $|S\rangle$, which means that the same qubits $n = 5$ are used with the Feature Encoding Ansatz. The depth of quantum circuit in Variational

[2] The last qubit is entangled with the first qubit to create a closed chain.

[3] One special point is that when $(i + j) \mod n = 0$, the second qubit is defined as the $n$th qubit.

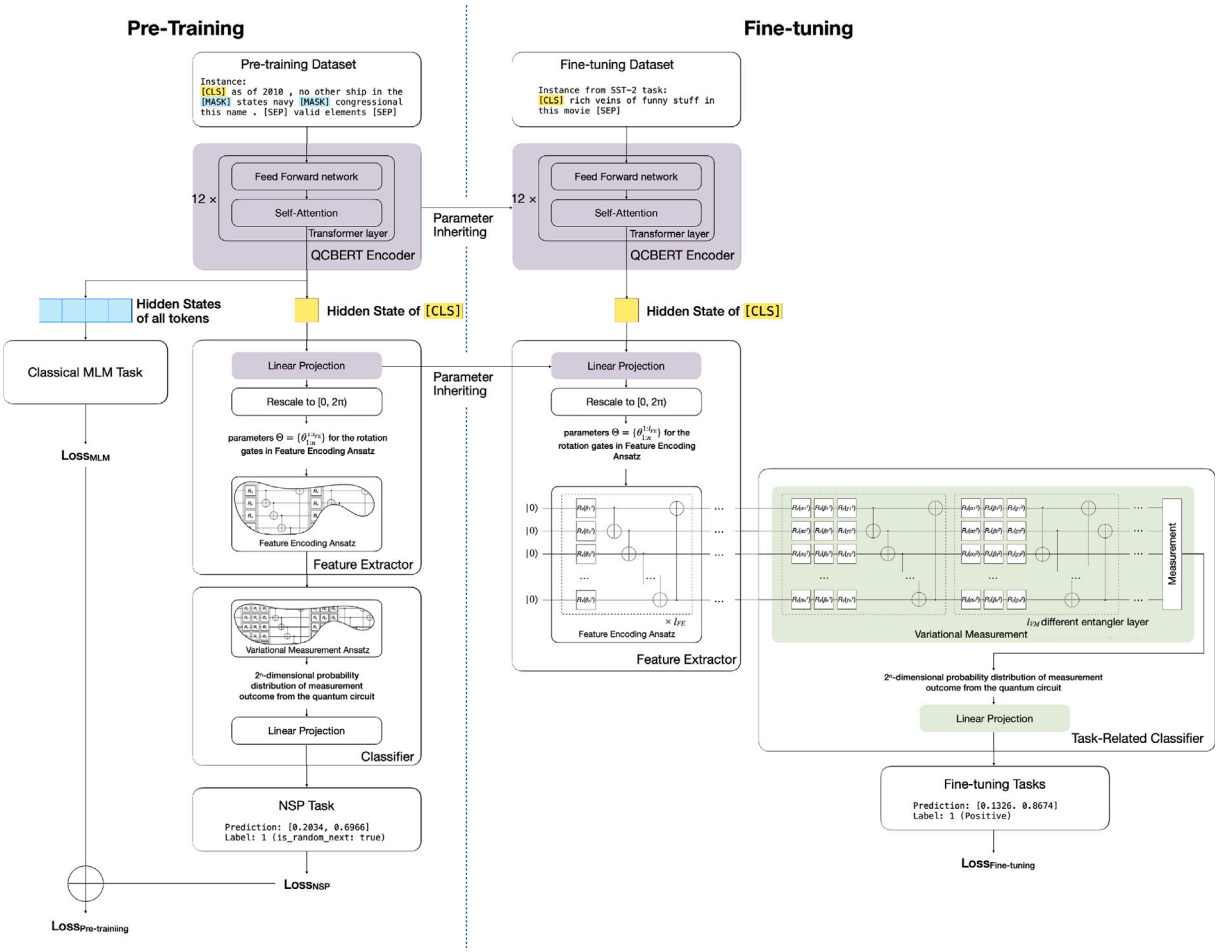



**Fig. 2.** The pipeline for pre-training and fine-tuning of QCBERT. During **pre-training** phase, the hidden states from the last Transformer layer of BERT encoder are utilized in the MLM task to get the loss, while the hidden state of [CLS] token is passed the QNLP architecture to implement the NSP task. The combined loss of these two tasks is then employed as the final loss for training model weights. During **fine-tuning**, the parameters of QCBERT encoder and linear projection layer before Feature Encoding Ansatz are inherited from the corresponding pre-trained parts of the model, and the parameters of task-related Variational Measurement Ansatz and its subsequent linear projection layer are initialized from scratch. In the figure, a sentence in the SST-2 is fed into the pre-trained QCBERT encoder to obtain the sentence representation, which is then projected and mapped to a quantum sentence encoding. Next, the task-related variational measurement circuit is applied to classify the sentence, where the measurement results are fed into a post-processing linear projection layer to yield the classification label.

Measurement Ansatz is a flexible choice, and it is set to $l_{VM} = 5$ in this work. This means the variational measurement circuit is parameterized by angles $\boldsymbol{\Phi} = \{A_{1:n}^{1:l_{VM}}, B_{1:n}^{1:l_{VM}}, \Gamma_{1:n}^{1:l_{VM}}\}$, with a parameter size of (5, 5, 3). Different from feature encoding, they are task-related and free trainable parameters that will be initialized and optimized for given tasks in the fine-tuning stage.

At the end of all quantum circuits, a measurement is applied to all $n$ qubits. The measurement outcomes, essentially a $2^n$-dimensional probability vector, are projected to the classification labels via a classical fully-connected layer. The parameters of this task-related linear projection layer are initialized and optimized in specific downstream tasks.

### 4.2. Pre-training

The whole model is jointly optimized at the pre-training stage. The self-supervised binary NSP task which is processed by the quantum classifier, while the self-supervised MLM task is handled classically. The quantum computing in the NSP task is conducted under simulating on a classical computer. The NSP task contains coherent sentence pairs for 50% of the pre-training examples and incoherent pairs by randomly sampling for the other 50%. For the MLM task, 15% of the token positions in pre-training data are randomly corrupted, in which 80% are masked, 10% are random tokens and the remaining 10% are unchanged.

The classical MLM loss and quantum NSP loss are summed up to produce the loss function. Classical backpropagation algorithm AdamW is used to update all model weights. Learning rate is set to 2e-5 and batch size is set to 64 per GPU.

The pre-training dataset is generated from the large text corpora BOOKCORPUS and English WIKIPEDIA (Devlin et al., 2019; Zhu et al., 2015). Two epochs of the dataset are generated for pre-training, and each is equally split into 100 chunks. Therefore, the whole pre-training procedure consists of 200 training epochs. The model with the lowest validation loss is saved to be fine-tuned in the subsequent downstream tasks.

### 4.3. Fine-tuning

The fine-tuning step is conducted under simulating the quantum computing on a classical computer. To explore the impact of noise in a real quantum run, we introduce a bit flip error with probability $p_{\text{flip}} = 1\%$ to each qubit in the circuit. The pre-trained QCBERT encoder is frozen, bringing a fixed hidden state for each input sample. The parameters of the linear projection layer in the feature extractor are employed from pre-trained values and set trainable in the fine-tuning

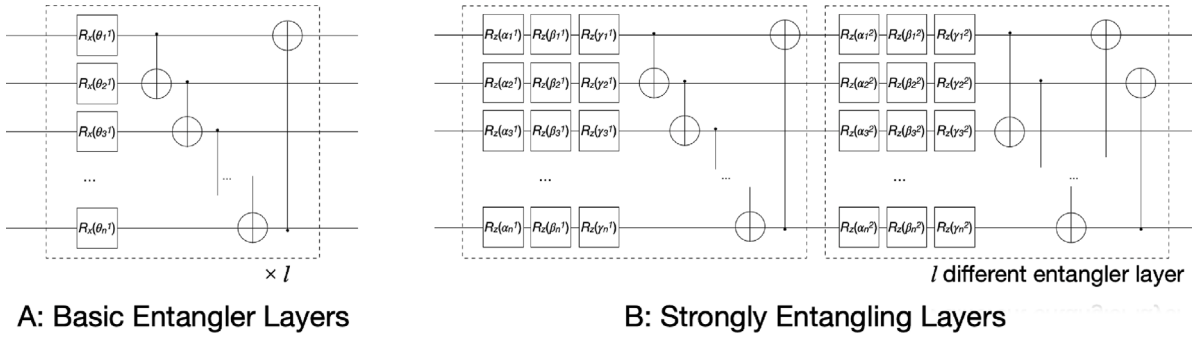



**Fig. 3.** The pre-built hardware-efficient quantum circuits which are used in QCBERT. A: the basic entangler layers in the Feature Encoding Ansatz. Each layer consists of one-parameter single-qubit rotations about the $x$ axis and *CNOT* gates which entangle each qubit with their neighbors. The same quantum circuit architecture will repeat $L$ times when the depth is set to be $L$. B: the strongly entangling layers in the Variational Measurement Ansatz. Each layer carries out single-qubit rotations about the $z$, $y$ and $z$ axes, and then entangles all $n$ qubits with ($n$) sequentially arranged *CNOT* gates.

phase. This enables the rotation angles in the Feature Encoding Ansatz to also benefit from pre-training and undergo further optimization in downstream tasks. The rotation angles in the Variational Measurement Ansatz are randomly initialized and optimized with the task-related data. The classical post-processing layer is also re-designed for particular classification task. Precisely, it contains a weight matrix $W_{task}$ of $2^n \times K$ for a k-class classification task, which is jointly updated in the fine-tuning stage.

In the experiment section, we fine-tuned QCBERT model on several datasets from the GLUE benchmark (Wang et al., 2018) to validate the effectiveness of our pre-training paradigm. Furthermore, we fine-tuned QCBERT on various small-scale datasets and compared its results with the QNLP model DisCoCat (Lorenz et al., 2021). Finally, we fine-tuned QCBERT on IAC-V1, IAC-V2 and the Twitter hate speech detection dataset to explore its generalizability and potential applications in real world. Both the fine-tuning and inference on the test samples were conducted via noisy simulation as mentioned above.

## 5. Experiment

In this section, we elaborate on the evaluation of our pre-trained QCBERT model. In the first part, our model was compared with the classical BERT on the GLUE benchmark to assess its absolute language understanding capability. In addition, we compared our model against the existing DisCoCat model to reveal the impact of self-supervised pre-training paradigm on a QNLP model. Notably, small-scale datasets were curated based on the GLUE datasets, and combined with the MC and RP datasets to serve as the benchmark for this purpose. Furthermore, we extended our QCBERT model to the tasks of sarcasm detection and Twitter hate speech detection, which are more practical sentiment-related tasks, to explore its potential applications and feasibility in real life. Finally, an ablation study was carried out, in which we investigated the impact of quantum circuit selection on QCBERT.

All of our experiments rely on simulating the quantum computing on a classical computer. The implementation of the quantum models is based on *Pytorch* and *Pennylane* (Bergholm et al., 2022), while the implementation of DisCoCat model depends on the *Lambeq* (Kartsaklis et al., 2021b) package. All models are pre-trained on the remote cluster using a Slurm scheduler. Each job requests a single node with 8 AMD Instinct MI250X GPUs and 100 GB of memory. They are fine-tuned in a single node with 1 NVIDIA A100.

### 5.1. Language understanding capability of QCBERT

A fair indicator of the capability of a pre-trained model is its performance in the downstream task. Therefore, we aimed to explore the competence of the self-supervised pre-trained QCBERT model by evaluating its performance in the GLUE benchmark, which covers different language understanding tasks and requires a high adaptability for a pre-trained language model. Meanwhile, we included another QCBERT variant in the comparison to showcase the benefits of pre-training to the quantum text representations.

#### 5.1.1. Datasets

The GLUE benchmark was utilized as the metric of our QCBERT models, which is a tool of assessing the natural language understanding capability of models. Due to its nature as a collection of various natural language tasks, it is considered to encourage cross-task sharing of general language knowledge (Wang et al., 2018). The MNLI, QNLI, RTE, SST-2, MRPC, CoLA and STS-B datasets were adopted. We employed the standard accuracy-based metrics for evaluating the models on these tasks, where MNLI, QNLI, RTE, and SST-2 tasks used *Accuracy*, the CoLA task used *Matthews Correlation Coefficient (MCC)*, and QQP and MRPC utilized the *F1* score due to their unbalanced class distributions. For the STS-B task, the *Spearman rank correlation coefficient* was applied to measure the correlation between the correct labels and predictions. The average of results of separated GLUE tasks was calculated as the GLUE score.

#### 5.1.2. Models

The following models were involved in the experiment:

***BERT-base.*** Our model differs from classical BERT in the way the NSP task is implemented. We would like to compare our model with BERT to assess the absolute language understanding capability of our model, and empirically demonstrate the influence of the unique quantum NSP structure. Therefore, we compared our QCBERT model with the classical model **BERT-base**. The entire BERT-base was trainable during fine-tuning.

***BERT-frozen-encoder.*** Considering that the QCBERT encoder in our model is frozen, we introduced a new comparison model, **BERT-frozen-encoder**, which had the same architecture and initial weights as BERT-base, but with a frozen encoder during fine-tuning. The pooling layer, responsible for generating the [CLS] token representation, and the final fully connected layer, acting as the classification head, were fine-tuned. The fine-tuning learning rate was set to be 5e-5, with 20 epochs and a batch size of 32. The best performance was measured by taking the checkpoint with the best corresponding metric values in different downstream tasks.

***QCBERT-base.*** In our model, the structure of Variational Measurement Ansatz in the task-related classifier is adjusted according to specific tasks. For a fair comparison with BERT-base, the encoder of our **QCBERT-base** had the same setup with 12 layers, 12 attention heads and dimensionalities of 768 and 3072 for hidden and intermediate layers. The quantum circuit has 5 qubits, 10 basic entangling layers in the pre-trained feature extractor and 5 strongly entangling layers

**Table 1**
Results of **QCBERT** on GLUE in comparison to other models. The evaluation metric for each dataset is shown in the parentheses below the dataset name. Performance values on the validation set are reported. For the MNLI task, the number before the slash refers to the result on the MNLI matched dataset and the number after the slash refers to the result on the MNLI mismatched dataset. Besides the BERT-base model, the results of the best-performing models in different downstream tasks are shown in bold.

| Model | GLUE benchmark | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | MNLI (m/mm) (Acc) | QNLI (Acc) | QQP (F1) | RTE (Acc) | SST-2 (Acc) | MRPC (F1) | CoLA (MCC) | STS-B (Spearman) | Avg (GLUE score) |
| BERT-base | 84.6/83.4 | 90.5 | 71.2 | 66.4 | 93.5 | 88.9 | 52.1 | 85.8 | 79.6 |
| BERT-frozen-encoder | 57.1/58.3 | 77.1 | **72.8** | 60.7 | **87.4** | 82.8 | **43.1** | 83.2 | 69.2 |
| QCBERT-base | **60.6/59.2** | **77.4** | 72.0 | **62.8** | 85.1 | **83.7** | 37.6 | **85.0** | **69.3** |
| QCBERT w/o pre-train | 40.2/38.8 | 74.2 | 70.8 | 57.0 | 85.0 | 81.6 | 36.6 | 80.7 | 62.8 |
| BERT-QChead | 37.7/28.9 | 74.8 | 71.1 | 58.9 | 87.0 | 83.3 | 39.2 | 80.8 | 62.4 |

in the task-related classifier. We investigated the effects of different quantum circuit ansatz on the performance of model in Section 5.4. In fine-tuning, the text encoder was frozen, and the parameters in the feature extractor and classifier were trainable. The fine-tuning learning rate was set to be 5e-3, with 20 epochs and a batch size of 32. The best performance was measured by taking the checkpoint with the best corresponding metric values in different downstream tasks.

***QCBERT w/o pre-train.*** We were interested in investigating the usefulness of pre-training for the quantum classification model. For this purpose, we implemented the feature extractor and classifier directly from the BERT-base encoder without pre-training, and fine-tuned the architecture in the same way as QCBERT. As an illustration of the positive impact of pre-training, QCBERT-base was expected to beat this **QCBERT w/o pre-train** variant on GLUE benchmark.

***BERT-QChead.*** In view of the distinction between the QNLP model architecture and classical classification network, we posited that the pre-trained QCBERT encoder was more adaptive to our QNLP model architecture over the pristine BERT encoder. To verify this assumption, we implemented another QCBERT variant **BERT-QChead**: a model with a **classical BERT-base encoder** and the **pre-trained QCBERT linear projection layer**. It was fine-tuned in the same way as QCBERT-base. We anticipated that our QCBERT-base model would surpass this variant in terms of the GLUE score. At the same time, due to the presence of pre-trained QCBERT, it was more adaptive to the quantum classification architecture than QCBERT w/o pre-train and expected to outperform it on GLUE tasks.

#### 5.1.3. Result

Table 1 shows the result of the experiment, in which the corresponding metrics are reported for all the GLUE datasets involved. The result of BERT-base are taken directly from its original paper (Devlin et al., 2019).

The benefit of self-supervised pre-training can be evidently observed by comparing QCBERT-base with QCBERT w/o pre-train. The trained text encoder and Feature Encoding Ansatz manage to project the text representations onto the parameters of the quantum circuit in the feature extractor, facilitating improved performance in downstream tasks. Compared to the end-to-end model (62.8 GLUE score), our pre-trained QCBERT-base model achieves superior performance across most GLUE tasks, yielding an overall 69.8 GLUE score. Furthermore, we further note that the QCBERT-base model outperforms BERT-QChead as expected with a remarkable gap of 6.9 on GLUE score. The performance gaps between the QCBERT-base and the two variants mainly stem from the MNLI task. Due to the absence or incompleteness of the self-supervised pre-training mechanism in either the QCBERT model without pre-training or the BERT-QChead model, they fail to handle the MNLI task that requires deeper semantic reasoning abilities and greater generalizability between various text styles and genres. This allows us to gain a clear understanding of the effect of our hybrid classical-quantum self-supervised pre-training mechanism, emphasizing the necessity of the joint pre-training of the classical encoder and quantum head.

We acknowledge that our model is still inferior to the classical BERT-base model (79.6 GLUE score), which has all modules being trainable, by a significant margin. Nevertheless, when compared to the BERT-frozen-encoder variant (69.2 GLUE score), in which the text encoder is also frozen to provide an aligned comparison to our own model, our QCBERT-base model demonstrates comparable performance. The results indicate that, when benefiting from a fixed and well pre-trained text representation, the quantum classification head has similar potential to the classical classification head in performing tasks that require advanced language understanding capabilities. Therefore, a feasible method to further improve our QCBERT-base model is the upgrading of text representation. Please refer to Section 6 for a more insightful discussion in this regard.

### 5.2. Benefit of QCBERT over existing QNLP models

We compared QCBERT with DisCoCat (Lorenz et al., 2021) to evaluate the performance gain brought about to the existing QNLP models. In addition to the MC (meaning classification) and RP (relative pronoun) tasks which were wildly-used by DisCoCat, we sampled partial data from the GLUE tasks to generate a series of small-scale datasets. We fine-tuned the pre-trained QCBERT model on these downstream tasks and contrasted it with the DisCoCat. Meanwhile, the QCBERT w/o pre-train variant was involved as a reference.

#### 5.2.1. Datasets

The MC dataset, RP dataset and the small-scale sampled GLUE datasets were employed to be the downstream fine-tuning tasks in this experiment.

In a series of studies related to DisCoCat-family models, MC and RP are widely used as metrics to test the performance of these pure quantum models. The MC task is a binary classification task for sentences, in which all the sentences consist of 3–4 words chosen from a vocabulary of 17 words and adhere to the syntactic structure of "noun phrase + verb phrase". The training set in the MC task is 70 and the size of both validation set and test set is 30. The RP dataset is a new dataset consists of 105 noun phrases containing relative clauses extracted from the RelPron dataset (Rimell, Maillard, Polajnar, & Clark, 2016) by Lorenz et al. (2021), in which the training set comprises 74 instances and the test set comprises 31 instances. The RP task is also a classification task, of which the objective is to determine whether the clause in the noun phrase is a subject relative clause or an object relative clause. Therefore, RP task requests the model to have a certain level of syntax sensitivity. Compared to the MC task, the RP task has a larger vocabulary, in which there are 115 words.

In addition, we included small-scale downstream tasks to increase the diversity of the experiment. Due to the high computational cost of syntactical analysis required by the DisCoCat model and the exponentially increasing demand for quantum computation resources with the length of sentences, it is challenging to use the DisCoCat model to process large-scale datasets containing many complex and lengthy sentences. To get the small-size dataset meeting the requirements, we filtrated the GLUE datasets according to specific criteria except the

**Table 2**
Results of **QCBERT** on sampled GLUE benchmark, MC and RP tasks in comparison to other models. The results of the best-performing models in different downstream tasks are highlighted in bold.

| Model | Sampled GLUE tasks | | | MC (Acc) | RP (Acc) |
|---|---|---|---|---|---|
| | QQP (F1) | SST-2 (Acc) | CoLA (MCC) | | |
| DisCoCat | 64.2 | 52.6 | 7.7 | 79.8 | 72.3 |
| QCBERT-base | **82.0** | **88.0** | **49.5** | **100.0** | **100.0** |
| QCBERT w/o pre-train | 81.2 | 84.0 | 44.7 | 100.0 | 96.8 |

MNLI and STS-B tasks. The criteria were: (I) The length of the sentence should be shorter than or equal to 8 without punctuation considered. (II) Rehearsing processing sentences according to the pipeline of DisCoCat model can yield meaningful results. For each GLUE task, we wanted to get total 350 instances which is class balanced. We only had sufficient data in the SST-2, CoLA, and QQP datasets, while the other datasets did not provide enough instances that met the criteria. We performed 5-fold cross-validation over all models for the three sampled GLUE tasks, considering their relatively small scale. In this process, the entire data sampled from the original training set and validation set was used.

#### 5.2.2. Models

In this experiment, we utilized both the QCBERT-base model and QCBERT w/o pre-train model. In view of the small size of all datasets in this experiment, the epoch of QCBERT variants was set to 50 to ensure sufficient training. Furthermore, we compared the results of the QCBERT-base model with those of the pure QNLP model DisCoCat.

*5.2.2.1.* ***DisCoCat***. DisCoCat model is a pure QNLP model which requires strict pre-processing of data to obtain the quantum circuits of sentences and trainable parameters in model. Although the pregroup grammar can simplify sentence structures and help to acquire concise quantum circuit representations, it introduces the risk of the model struggling to handle complex sentences. In our proposed QCBERT models, using text representation to generate fixed-size quantum circuits can effectively avoid this issue while benefiting from pre-training. To provide a more intuitive comparison between our pre-trained model and the pure QNLP model, we selected and trained the DisCoCat model on the datasets. For the DisCoCat model, the results of the MC and RP task directly referenced the study by Lorenz et al. (2021). Meanwhile, to get the result of the sampled GLUE tasks from DisCoCat, we adopted a batch size of 100, an epoch of 20, and a shot of 10000 for measurement after each quantum circuit.

#### 5.2.3. Result

Table 2 exhibits the primary result of the experiment, in which all the sampled GLUE tasks report the corresponding metrics. For the MC and RP tasks, accuracy is reported as the measuring standard. We first compare the result of the QCBERT-base model with the result of the QCBERT w/o pre-train model. Maintaining consistency with the result of the previous experiment, the pre-trained model exhibits comprehensive advantages in terms of capability, compared to the end-to-end model. However, for the MC and RP datasets, the gap in performance is less pronounced due to the relative simplicity of these tasks, which lead to a performance ceiling.

Comparing the result of the QCBERT-base model with the result of the DisCoCat model, we find that the pre-trained model shows superiority over the DisCoCat model across all the tasks. The disparity in performance is evident on the SST-2 and CoLA tasks, where the complexity of vocabulary of these two tasks surpasses those of simple MC and RP models. In addition to its performance advantage over the DiaCoCat model, the QCBERT-base model also proves to be more cost-effective. Specifically, taking the sampled QQP task as an example, the DisCoCat model takes **87.4** times longer than the QCBERT-base model to complete the training and evaluation of the dataset.

Apart from analyzing the results of models on these downstream tasks, it is also worthwhile to analyze the results of sampling from the GLUE benchmark when creating small-scale datasets. We successfully sampled enough data only in three tasks, exhibiting the high demands of DisCoCat on data quality and its low generality.

**Table 3**
The average length of sentences across various tasks under different scenarios in the case study. The row name $\overline{L}_1$ refers to the average length of all sentences predicted correctly by the corresponding model. $\overline{L}_0$ refers to the average length of all sentences predicted wrongly by the corresponding model is calculated. $\overline{L}_>$ refers to the average length of the samples where the corresponding model (the sub-column name) predicts correctly while the another model makes a wrong prediction.

| Name | QCBERT-base | DisCoCat |
|---|---|---|
| $\overline{L}_1$ | 4.58 | 4.37 |
| $\overline{L}_0$ | 4.36 | 4.38 |
| $\overline{L}_>$ | 4.31 | 4.05 |

**Table 4**
Case Study. Sentences sampled from the SST-2 dataset are predicted correctly by the QCBERT-base but wrongly by the DisCoCat model. Label 1 indicates that the sentence is *positive*, while label 0 means the sentence is *negative*.

| Example Sentences | Golden | $\text{Pred}_{\text{DisCoCat}}$ | $\text{Pred}_{\text{QCBERT}}$ |
|---|---|---|---|
| *a wonderful , sobering , heart-felt drama* | 1 | 0 | 1 |
| *evans ) had , lost , and got back* | 1 | 0 | 1 |
| *hopeful and , perhaps paradoxically , illumina...* | 1 | 0 | 1 |
| *better characters , some genuine quirkiness and* | 1 | 0 | 1 |
| *( a ) thoughtful , visually graceful* | 1 | 0 | 1 |
| *: it 's not really funny .* | 0 | 1 | 0 |
| *essentially ruined – or , rather , overpowered* | 0 | 1 | 0 |
| *the action clichés just pile up .* | 0 | 1 | 0 |

#### 5.2.4. Case study

In order to better illustrate the advantages of our QCBERT architecture over the pure QNLP model and to elucidate where these advantages stem from, we analyzed the features of the dataset and attempted to provide a preliminary answer to this question by this case study.

DisCoCat heavily depends on sentence analysis based on pregroup grammar for deriving subsequent quantum circuits and trainable parameters within the model. Given that longer sentences typically imply more intricate sentence structures, we investigate the average sentence length across varying situations where the QCBERT-base model and DisCoCat model perform differently, to explore the potential relationship between model performance and sentence complexity.

We first compute the average sentence length in the circumstances where the corresponding model predicts correctly, acquiring the $\overline{L}_1$ for the small-scale sampled SST-2 task. The sampled SST-2 dataset is adopted in the case study because its original task contains abundant samples, and we consider it to be representative. Furthermore, the $\overline{L}_0$ of sentences which are predicted wrongly by the models is calculated. The results are exhibited in Table 3. We find that the QCBERT-base model exhibits a greater ability to deal longer sentences, as the $\overline{L}_1 = 4.58$ has a significant preponderance than the $\overline{L}_0 = 4.36$ in this model, which is not observed for the DisCoCat model.

To further differentiate the capabilities of the QCBERT-base model and the DisCoCat model in handling data with different characteristics, we employ another metric, $\overline{L}_>$. This metric indicates the average length of sentences predicted correctly by one model but wrongly by the other. For instance, the $\overline{L}_> = 4.31$ signifies a scenario where the QCBERT-base model yields successful predictions, whereas the DisCoCat model

**Table 5**
Results of **QCBERT** on the IAC-V1, IAC-V2 and Twitter hate speech detection tasks in comparison to other models. The results of the best-performing models in different downstream tasks are highlighted in bold.

| Model | IAC-V1 | | IAC-V2 | | Twitter hate | | Avg. of Ma-F1 |
|---|---|---|---|---|---|---|---|
| | Acc | Ma-F1 | Acc | Ma-F1 | Acc | Ma-F1 | |
| BERT-frozen-encoder | 66.3 | 66.2 | 75.9 | 75.9 | **95.6** | **82.3** | **74.8** |
| QCBERT-base | **66.9** | **66.8** | **76.5** | **76.5** | 92.8 | 77.8 | 73.7 |
| QCBERT w/o pre-train | 63.4 | 63.2 | 75.5 | 75.5 | 88.4 | 72.1 | 70.3 |

fails. The result in the bottom half of Table 3 demonstrates an obvious superiority of our self-supervised pre-trained model ($\overline{L}_{>} = 4.31$) over the pure QNLP model ($\overline{L}_{>} = 4.05$) in handling more complex sentences.

We further showcase several typical sentences from the sampled SST-2 dataset, which are correctly predicted by the QCBERT-base model but wrongly by the DisCoCat model.

These eight sentences in Table 4 characterize common patterns that lead to wrong predictions by DisCoCat. Rather than samples in the MC and RP dataset, real sentences do not strictly follow the typical language structures, and sometimes even have incomplete structures. Furthermore, the presence of compound adjectives and contractions in these sentences undoubtedly increases the complexity of the sentences and the cost required for sentence structure analysis. Both factors potentially contribute to the failure of DisCoCat. In contrast, our QCBERT architecture handles these data more adeptly, relying more on the sentence semantics, thereby yielding correct predictions for these sentences.

### 5.3. Real-world applications of QCBERT

Apart from its natural language understanding capability, another key indicator for evaluating the QCBERT model is assessing its generalizability and potential real-world applications. In Section 5.1, we have preliminarily discussed the generalizability of the pre-trained QCBERT-base model by analyzing the results of MNLI task in different models. Nevertheless, the most straightforward approach is to evaluate the QCBERT-base model on tasks beyond the scope of its pre-training capabilities. Considering that sentiment reasoning was not a primary objective in the pre-training phase of the QCBERT-base model, in this section we intuitively explore the generalizability and potential real-world applicability of our model by introducing sentiment classification tasks.

#### 5.3.1. Datasets

The sarcasm detection tasks sourced from Internet Argument Corpus 1.0 (IAC-V1) (Lukin & Walker, 2017) and Internet Argument Corpus 2.0 (IAC-V2) (Oraby et al., 2017), and the Twitter Hate speech Detection dataset[4] were employed in this experiment.

IAC-V1 and IAC-V2 used in the experiment are both dialogue-based datasets that align with the frequently dialogic environment of social media. Meanwhile, the Twitter Hate speech Detection dataset is a collection of tweets which consists of real, user-generated speeches on social media platforms. Furthermore, the quantum NSP task and classical MLM task are primarily designed to help the QCBERT-base model learn general language understanding rather than specific sentiment reasoning. Based on these factors, we believed that these three tasks were able to test the generalizability of the QCBERT-base model, and explore its feasible applications in the real world.

#### 5.3.2. Models

In this experiment, we used both the QCBERT-base model and QCBERT w/o pre-train model with a learning rate of 5e-3. Besides, the BERT-frozen-encoder variant was also included to work as a comparison with a learning rate of 5e-5. For the three models, the epoch was set to 20.

[4] https://www.kaggle.com/datasets/vkrahul/twitter-hate-speech

#### 5.3.3. Result

Table 5 exhibits the results on the IAC-V1, IAC-V2 and Twitter hate speech detection tasks from the QCBERT-base, QCBERT w/o pre-train and BERT-frozen-encoder models.

Compared to the BERT-frozen-encoder model, our QCBERT-base model demonstrates comparable performance on the two sarcasm detection tasks, which implies its potential to handle similar tasks. However, our model displays evident inferiority in Twitter hate speech detection, resulting in an overall statistical lag compared to the BERT-frozen-encoder model. On the one hand, the suboptimal performance potentially results from the weaker capability for processing information that contains more noise, considering the hashtags, emojis, and other distracting information in tweets. On the other hand, the unbalanced label classes in the Twitter hate speech detection task may cause the relative failure for the QCBERT-base model to learn meaningful features. In summary, our QCBERT-base model truly exhibits an ability to handle sentiment-related tasks, showcasing language processing capabilities that extended beyond its pre-training, thus implying its generalizability. Furthermore, these results also reveal the potential applications of our model in real-world tasks.

### 5.4. Ablation study: Impact of quantum circuit ansatz to QCBERT

When designing the QCBERT model architecture, a key issue to focus on is the selection of hyper-parameters in the quantum part, which is closely related to the trade-off between the performance and the computational costs of our model. Although more parameters may lead to better model performance, they also increase the time consumption, a problem that is particularly pronounced in quantum models. Specifically, this involves deciding on the structures for the two quantum circuits in the feature extractor and the task-related classifier. In this section, we examined the impact and computational costs of different quantum circuit ansatz across various QCBERT modules.

#### 5.4.1. Settings

There were three independent variables in this ablation study: (I) the number of qubits used in the quantum circuit; (II) the depth of the quantum circuit in the feature extractor during pre-training; (III) and the depth of the quantum circuit in the task-related classifier during fine-tuning. During the pre-training phase, we trained models with different hyper-parameters, where the number of qubits was set to 5/10, and the depth of the feature encoding circuit was set to 5/10/15/20. Subsequently, in the downstream tasks, we changed the depth of the quantum circuit in the variational measurement circuit, and fine-tuned these models.

All models in this section were constructed based on the QCBERT architecture. During the pre-training phase, for all the QCBERT models, only the number of qubits and the depth of the quantum circuit in the Feature Encoding Ansatz varied; all other hyper-parameters remained constant. For all models, during the pre-training, the optimization algorithm was set to AdamW, and a learning rate of 2e-5 with a batch size of 64 were employed. The number of pre-training epochs was set to 200, indicating that the entire dataset was traversed through twice.

During the fine-tuning phase, we adjusted the depth of the variational measurement circuits. For models which were pre-trained in various conditions, we fine-tuned them on four different scenarios: The depth $l_{VM}$ of the strongly entangling layers was set to 5/10/15/20.

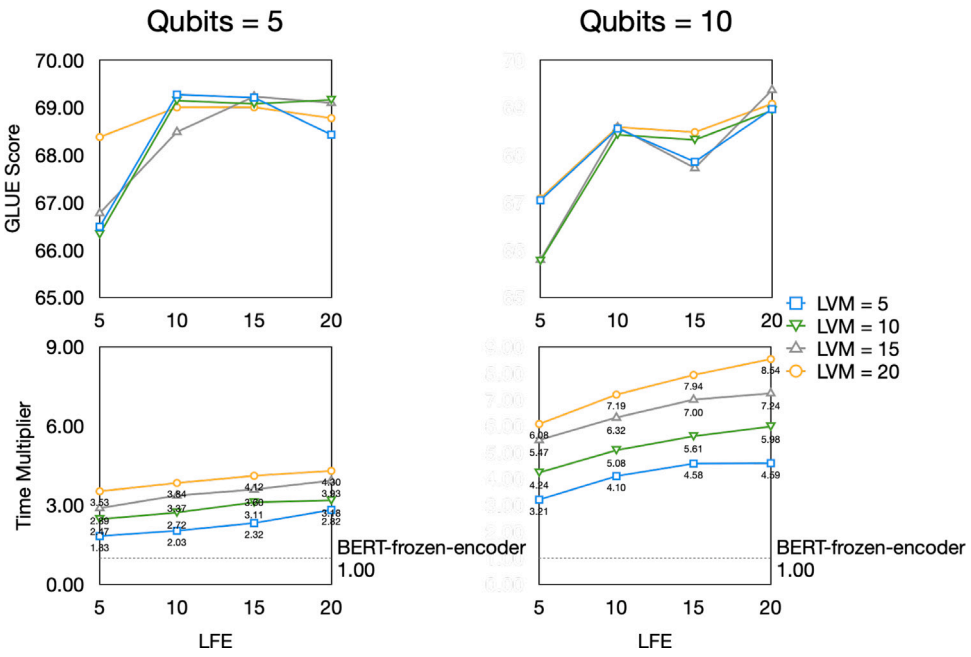



**Fig. 4.** Results of the ablation study to explore the impact of quantum circuit ansatz to QCBERT. In the top sub-figures, the average outputs across all GLUE tasks for all models are reported. In the bottom sub-figures, the time multipliers compared to the BERT-frozen-encoder model are shown. The x-axis label $l_{FE}$ refers to the depth of quantum circuit in the Feature Encoding Ansatz, which employs basic entangler layers. The category $l_{VM}$ refers to the depth of quantum circuit in the Variational Measurement Ansatz, which employs strongly entangling layers.

We maintained consistency in the other hyper-parameters across all models. In the process of fine-tuning, we froze the parameters in QCBERT encoders, and employed a learning rate of 5e-3, as well as a batch size of 32. The number of epochs was set to 20 for all models.

### 5.4.2. Datasets

During the pre-training phase, the dataset generated from BOOKCORPUS and English WIKIPEDIA was utilized to pre-train the models. During the fine-tuning tasks, the Dataset used was also the GLUE benchmark. These tasks, including MNLI, QNLI, QQP, RTE, SST-2, MRPC, CoLA, and STS-B, were performed across all pre-trained models. The final GLUE score was computed by averaging all the results of tasks.

### 5.4.3. Result

Fig. 4 exhibits the result of this ablation study. The top sub-figures showcase the performance of model, while the bottom sub-figures display the computational costs. The column title *Qubits* explains the number of qubits $n$. The x-axis label $l_{FE}$ states the depth of feature encoding circuit in the pre-trained feature extractor. And the category $l_{VM}$ clarifies the depth of quantum circuit employed in the Variational Measurement Ansatz for whole models. All of these three independent variables are the features we want to explore in this study.

We first compare the degrees of impact of the three independent variables. The results evidently showcase that the varying depths of the quantum circuit in the feature extractor have a more profound impact on the QCBERT models. For all the scenarios where $n$ is set to 5/10 and $l_{VM}$ is set to 5/10/15/20, the changes in $l_{FE}$ have a noticeable impact on the pre-training process. The models with the deeper length of the quantum circuit in the feature extractor gain enhanced capability in processing textual information through pre-training, resulting in significantly superior performances in subsequent downstream tasks. However, the benefit of increasing $l_{FE}$ reaches a plateau. All models with $l_{FE} \geq 10$ produce similar outputs.

The impact of $l_{VM}$ on the performance of the QCBERT model is limited and irregular. Smaller $l_{VM}$ values do not always lead to worse performance. For the independent variable $n$, at least in this work, the number of qubits in the quantum circuit has restricted influence to the capability of QCBERT model, which is beyond our expectation. Although the number of trainable parameters in the models with $n = 5$ are significantly less than that with $n = 10$, their performance is still comparable to the latter. Specifically, the best performance of QCBERT model, with a GLUE score of 69.4, comes from the scenario where $n = 10$, $l_{FE} = 20$ and $l_{FE} = 15$. The second-best GLUE score of 69.3 is achieved in the scenario where $n = 5$, $l_{FE} = 10$ and $l_{FE} = 5$, with an exponential reduction in computing costs.

It is crucial to consider the computational costs of the models while evaluating its performance. According to the time multipliers shown in the bottom sub-figures, the rising values of $n$ and $l_{VM}$ result in exponential increases in computational time, even though they do not lead to improvements in model performance. In comparison to the BERT-frozen-encoder model, the best-performing QCBERT model ($n = 10$, $l_{FE} = 20$ and $l_{FE} = 15$) requires 7.24 times the computational cost, while the second-best QCBERT model ($n = 5$, $l_{FE} = 10$ and $l_{FE} = 5$) only needs 2.03 times.

Based on the above results and a comprehensive consideration of the trade-off between model performance and computational costs, we select the model with 5 qubits, 10-depth basic entangler layers in the feature extractor, and 5-depth strongly entangling layers in the task-related classifier as the QCBERT-base model for the main experiments.

## 6. Discussion

The well-designed self-supervised pre-training paradigm indeed enhances the capability of the QNLP models. The effect of the pre-training mechanism is primarily reflected in two aspects. First, it enables the pre-trained text representations to better align with QNLP tasks. Second, it allows the parameters in the feature extractor to be sufficiently trained using a large amount of textual information. These advantages gained through pre-training are preserved in subsequent fine-tuning, giving rise to enhanced effectiveness in downstream tasks.

To a certain extent, our pre-trained quantum model overcomes the limitations in the representation capacity of quantum computing models, which stem from the high linearity of the underlying quantum

computing architecture. Specifically, our model offers several advantages over previous research in various aspects. On the one hand, our work fills the gap in the QNLP field, particularly in the research using self-supervised pre-trained QNLP models, illuminating the possibility of implementing similar paradigms or model architectures. On the other hand, compared to the pure QNLP model, our model demonstrates higher adaptability to a massive data containing long and difficult sentences with intricate structures, which is supported by the results of our case study analysis in Section 5.2.4. In the case study, we found that the QCBERT-base model could handle sentences with longer average lengths than DisCoCat.

Although the self-supervised pre-training method has achieved promising results in training QNLP models, some issues still cannot be overlooked. We provide insights into two main issues facing the current approach as follows.

### 6.1. Barren plateaus in training

As shown in Section 5.4, increasing the number of qubits and the depth of the quantum circuit does not lead to better performance no matter the rise in trainable parameters. Specifically, Only the shift from $l_{FE} = 5$ to $l_{FE} = 10$ stimulates a significant improvement in the performance of model. A similar pattern can be observed in both scenarios with 5 or 10 qubits. This suggests that QCBERT model might fall into a dilemma called *barren plateaus*.

The barren plateau landscapes are explained as a phenomenon in which the gradients vanish exponentially during the training of quantum neural networks (McClean, Boixo, Smelyanskiy, Babbush, & Neven, 2018; Pesah et al., 2021). The barren plateaus come from the excessive entanglements between qubits (Ortiz Marrero, Kieferová, & Wiebe, 2021), which precisely refer to the substantial presence of *CNOT* gates in our quantum circuits. In the ablation study where we wanted to explore the impact of quantum circuit ansatz to QCBERT, we found that the transition from a 10-depth to a 20-depth quantum circuits in the feature extractor exhibited a plateau in performance. This result is likely due to the conflict between the advantages provided by the increased number of parameters with increasing depth, and the risk of barren plateaus caused by the corresponding increase in entanglements. This result poses the latent upper limit of the QCBERT model resulting from the barren plateaus.

In addition to the excessive depth, the formation of barren plateaus in our model may also be pertinent to the choice of quantum circuit types in the feature extractor. We practically utilize the hardware-efficient ansatz in the model for compatibility with real quantum hardware. The basic entangling layer structure used in the feature extractor only contains one type of rotation gate ($R_x$ gate) in each layer, causing a homogeneous structure and limiting the optimization of parameters in the quantum circuits. To overcome the shortages of current model architecture, more elaborate quantum circuit structures can be introduced into our model in the future study. For instance, the data re-uploading circuit structure, in which the Feature Encoding Ansatz and Variational Measurement Ansatz alternate in a regular pattern (Jerbi et al., 2023), can potentially be an ideal structure for learning more meaningful and representative quantum text encodings.

### 6.2. Fine-grained text representations

The pure QNLP model DisCoCat relies on pregroup grammar knowledge to pre-process sentences before converting them into quantum ansatz, which naturally embeds all semantic, grammatical, and syntactic information within the quantum circuits. In our model, the text representations of the [CLS] token are adopted to handle the classification tasks, given that the [CLS] token is specifically designed for this type of task (Devlin et al., 2019). However, compared to the DisCoCat model, our current strategy overlooks the fine-grained textual information contained within all token representations. In our next research endeavor, incorporating all token representations in a reasonable manner will be a promising avenue to enhance the capability of our model. Some explorations have been carried out on a quantum self-attention mechanism, drawing on the principles of the classical self-attention mechanism (Vaswani et al., 2017). The hybrid quantum–classical model established through this method has been validated for its effectiveness (Li et al., 2024), which paves the way towards this direction for future work.

## 7. Conclusion

This work presents a self-supervised pre-training model architecture which is employed in the QNLP field and achieves a promising performance. The linguistic knowledge acquired through pre-training improves the capability of the pre-trained model over the end-to-end model, and assists quantum computing models in finding a potential path to break free from the constraints of linearity inherent in the quantum circuit itself. Simultaneously, pre-training on a massive data under the application of a hybrid quantum–classical model helps to generate suitable text representations for the QNLP model structure. Nevertheless, our model faces the barren plateaus problem and the inadequate application of text representations, and we plan to address these issues in future research to achieve a more powerful QNLP model.

## CRediT authorship contribution statement

**Ben Yao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Prayag Tiwari:** Writing – review & editing, Validation. **Qiuchi Li:** Writing – review & editing, Supervision, Methodology, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

## References

Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., et al. (2023). Gpt-4 technical report. arXiv preprint arXiv:2303.08774.

Bergholm, V., Izaac, J., Schuld, M., Gogolin, C., Ahmed, S., Ajith, V., et al. (2022). PennyLane: Automatic differentiation of hybrid quantum-classical computations. arXiv:1811.04968.

Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., & Lloyd, S. (2017). Quantum machine learning. *Nature*, *549*(7671), 195–202. http://dx.doi.org/10.1038/nature23474, https://www.nature.com/articles/nature23474.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., et al. (2020). In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, H. Lin (Eds.), *Advances in neural information processing systems, Vol. 33* (pp. 1877–1901). Curran Associates, Inc., https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.

Cao, Y., Romero, J., Olson, J. P., Degroote, M., Johnson, P. D., Kieferová, M., et al. (2019). Quantum chemistry in the age of quantum computing. *Chemical Reviews*, *119*, 10856–10915. http://dx.doi.org/10.1021/acs.chemrev.8b00803.

Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *International conference on machine learning* (pp. 1597–1607). PMLR.

Coecke, B., de Felice, G., Meichanetzidis, K., & Toumi, A. (2020). Foundations for near-term quantum natural language processing. arXiv:2012.03755.

Coecke, B., & Kissinger, A. (2018). Picturing quantum processes: A first course on quantum theory and diagrammatic reasoning. In *Diagrammatic representation and inference: 10th international conference, diagrams 2018, edinburgh, UK, June 18-22, 2018, proceedings 10* (pp. 28–31). Springer.

Coecke, B., Sadrzadeh, M., & Clark, S. J. (2010). Mathematical foundations for a compositional distributional model of meaning. *Linguistic Analysis*, *36*(1), 345–384.

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., et al. (2019). Unsupervised cross-lingual representation learning at scale. arXiv preprint arXiv:1911.02116.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. N. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)* (pp. 4171–4186).

Di Sipio, R., Huang, J.-H., Chen, S. Y.-C., Mangini, S., & Worring, M. (2021). The dawn of quantum natural language processing. http://dx.doi.org/10.48550/arXiv.2110.06510 Number: arXiv:2110.06510 [quant-ph].

Di Sipio, R., Huang, J.-H., Chen, S. Y.-C., Mangini, S., & Worring, M. (2022). The dawn of quantum natural language processing. In *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing* (pp. 8612–8616). IEEE.

Guarasci, R., De Pietro, G., & Esposito, M. (2022). Quantum natural language processing: Challenges and opportunities. *Applied Sciences*, *12*(11), http://dx.doi.org/10.3390/app12115651, https://www.mdpi.com/2076-3417/12/11/5651.

He, Z., Deng, M., Zheng, S., Li, L., & Situ, H. (2023). Gsqas: graph self-supervised quantum architecture search. *Physica A. Statistical Mechanics and its Applications*, *630*, Article 129286.

He, K., Fan, H., Wu, Y., Xie, S., & Girshick, R. (2020). Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9729–9738).

Jaderberg, B., Anderson, L. W., Xie, W., Albanie, S., Kiffner, M., & Jaksch, D. (2022). Quantum self-supervised learning. *Quantum Science and Technology*, *7*(3), Article 035005.

Jerbi, S., Fiderer, L. J., Poulsen Nautrup, H., Kübler, J. M., Briegel, H. J., & Dunjko, V. (2023). Quantum machine learning beyond kernel methods. *Nature Communications*, *14*(1), 517.

Kartsaklis, D., Fan, I., Yeung, R., Pearson, A., Lorenz, R., Toumi, A., et al. (2021a). *lambeq: An Efficient High-Level Python Library for Quantum NLP*: *Technical Report*, http://dx.doi.org/10.48550/arXiv.2110.04236, arXiv:2110.04236 [quant-ph] type: article.

Kartsaklis, D., Fan, I., Yeung, R., Pearson, A., Lorenz, R., Toumi, A., et al. (2021b). Lambeq: An efficient high-level python library for quantum NLP. arXiv preprint arXiv:2110.04236.

Konar, D., Bhattacharyya, S., Dey, S., & Panigrahi, B. K. (2019). Opti-QIBDS net: A quantum-inspired optimized bi-directional self-supervised neural network architecture for automatic brain MR image segmentation. In *TENCON 2019-2019 IEEE region 10 conference* (pp. 761–766). IEEE.

Konar, D., Bhattacharyya, S., Gandhi, T. K., & Panigrahi, B. K. (2020). A quantum-inspired self-supervised network model for automatic segmentation of brain MR images. *Applied Soft Computing*, *93*, Article 106348.

Konar, D., Bhattacharyya, S., Gandhi, T. K., Panigrahi, B. K., & Jiang, R. (2023). 3-D quantum-inspired self-supervised tensor network for volumetric segmentation of medical images. *IEEE Transactions on Neural Networks and Learning Systems*.

Konar, D., Bhattacharyya, S., Panigrahi, B. K., & Behrman, E. C. (2021). Qutrit-inspired fully self-supervised shallow quantum learning network for brain tumor segmentation. *IEEE Transactions on Neural Networks and Learning Systems*, *33*(11), 6331–6345.

Lambek, J. (1958). The mathematics of sentence structure. *American Mathematical Monthly*, *65*(3), 154–170.

Lambek, J. (1999). Type grammar revisited. In *Logical aspects of computational linguistics: second international conference, LACL'97 nancy, France, September 22-24, 1997 selected papers 2* (pp. 1–27). Springer.

Li, Q., Wang, B., Zhu, Y., Lioma, C., & Liu, Q. (2023). Adapting pre-trained language models for quantum natural language processing. arXiv:2302.13812.

Li, G., Zhao, X., & Wang, X. (2024). Quantum self-attention neural networks for text classification. *Science China. Information Sciences*, *67*(4), 1–13.

Liao, Y., Hsieh, M.-H., & Ferrie, C. (2021). Quantum optimization for training quantum neural networks. http://dx.doi.org/10.48550/arXiv.2103.17047, arXiv:2103.17047 [quant-ph].

Liu, Y., Li, Q., Wang, B., Zhang, Y., & Song, D. (2023). A survey of quantum-cognitively inspired sentiment analysis models. *ACM Computing Surveys*, *56*(1), http://dx.doi.org/10.1145/3604550.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., et al. (2019). Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692.

Liu, X., Zhang, F., Hou, Z., Mian, L., Wang, Z., Zhang, J., et al. (2021). Self-supervised learning: Generative or contrastive. *IEEE Transactions on Knowledge and Data Engineering*, *35*(1), 857–876.

Lloyd, S., Schuld, M., Ijaz, A., Izaac, J., & Killoran, N. (2020). Quantum embeddin gs for machine learning. arXiv preprint arXiv:2001.03622.

Lorenz, R., Pearson, A., Meichanetzidis, K., Kartsaklis, D., & Coecke, B. (2021). QNLP in practice: Running compositional models of meaning on a quantum computer. arXiv preprint arXiv:2102.12846.

Lukin, S., & Walker, M. (2017). Really? Well. Apparently bootstrapping improves the performance of sarcasm and nastiness classifiers for online dialogue. arXiv:1708.08572.

Mari, A., Bromley, T. R., Izaac, J., Schuld, M., & Killoran, N. (2020). Transfer learning in hybrid classical-quantum neural networks. *Quantum*, *4*, 340. http://dx.doi.org/10.22331/q-2020-10-09-340.

Mavroeidis, V., Vishi, K., Mateusz, D., & Jøsang, A. (2018). The impact of quantum computing on present cryptography. *International Journal of Advanced Computer Science and Applications*, *9*(3), http://dx.doi.org/10.14569/ijacsa.2018.090354, http://dx.doi.org/10.14569/IJACSA.2018.090354.

McArdle, S., Endo, S., Aspuru-Guzik, A., Benjamin, S. C., & Yuan, X. (2020). Quantum computational chemistry. *Reviews of Modern Physics*, *92*, Article 015003. http://dx.doi.org/10.1103/RevModPhys.92.015003, https://link.aps.org/doi/10.1103/RevModPhys.92.015003.

McClean, J. R., Boixo, S., Smelyanskiy, V. N., Babbush, R., & Neven, H. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Communications*, *9*(1), 4812.

Meichanetzidis, K., Gogioso, S., Felice, G. D., Chiappori, N., Toumi, A., & Coecke, B. (2020). Quantum natural language processing on near-term quantum computers. arXiv:2005.04147.

Mogalapalli, H., Abburi, M., Nithya, B., & Bandreddi, S. K. V. (2021). Classical–quantum transfer learning for image classification. *SN Computer Science*, *3*(1), 20. http://dx.doi.org/10.1007/s42979-021-00888-y.

Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information*. Cambridge University Press.

Oh, S., Choi, J., & Kim, J. (2020). *A tutorial on quantum convolutional neural networks (QCNN)*: *Technical Report*, arXiv, http://dx.doi.org/10.48550/arXiv.2009.09423, arXiv:2009.09423 [quant-ph] type: article.

Oraby, S., Harrison, V., Reed, L., Hernandez, E., Riloff, E., & Walker, M. (2017). Creating and characterizing a diverse corpus of sarcasm in dialogue. arXiv:1709.05404.

Ortiz Marrero, C., Kieferová, M., & Wiebe, N. (2021). Entanglement-induced barren plateaus. *PRX Quantum*, *2*, Article 040316. http://dx.doi.org/10.1103/PRXQuantum.2.040316, URL https://link.aps.org/doi/10.1103/PRXQuantum.2.040316.

Pandey, S., Basisth, N. J., Sachan, T., Kumari, N., & Pakray, P. (2023). Quantum machine learning for natural language processing application. *Physica A. Statistical Mechanics and its Applications*, *627*, Article 129123.

Pesah, A., Cerezo, M., Wang, S., Volkoff, T., Sornborger, A. T., & Coles, P. J. (2021). Absence of barren plateaus in quantum convolutional neural networks. *Physical Review X*, *11*, Article 041011. http://dx.doi.org/10.1103/PhysRevX.11.041011, https://link.aps.org/doi/10.1103/PhysRevX.11.041011.

Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum*, *2*, 79. http://dx.doi.org/10.22331/q-2018-08-06-79, arXiv:1801.00862 [cond-mat, physics:quant-ph].

Proos, J., & Zalka, C. (2004). Shor's discrete logarithm quantum algorithm for elliptic curves. arXiv:quant-ph/0301141.

Qi, J., Yang, C.-H. H., Chen, P.-Y., & Hsieh, M.-H. (2023). Pre-training tensor-train networks facilitates machine learning with variational quantum circuits. http://dx.doi.org/10.48550/arXiv.2306.03741, arXiv:2306.03741 [quant-ph].

Radford, A., & Sutskever, I. (2018). Improving language understanding by generative pre-training. In *Arxiv*.

Rani, V., Nabi, S. T., Kumar, M., Mittal, A., & Kumar, K. (2023). Self-supervised learning: A succinct review. *Archives of Computational Methods in Engineering*, *30*(4), 2761–2775.

Rimell, L., Maillard, J., Polajnar, T., & Clark, S. (2016). RELPRON: A relative clause evaluation data set for compositional distributional semantics. *Computational Linguistics*, *42*(4), 661–701.

Schuld, M., Bocharov, A., Svore, K. M., & Wiebe, N. (2020). Circuit-centric quantum classifiers. *Physical Review A*, *101*(3), http://dx.doi.org/10.1103/physreva.101.032308.

Schuld, M., & Petruccione, F. (2021). Quantum Models as Kernel Methods. In M. Schuld, & F. Petruccione (Eds.), *Quantum science and technology*, *Machine learning with quantum computers* (pp. 217–245). Cham: Springer International Publishing, http://dx.doi.org/10.1007/978-3-030-83098-4_6.

Suneel, S., Krishnamoorthy, R., Gopatoti, A., Maguluri, L. P., Kuncha, P., & Sunil, G. (2024). Enhanced image diagnosing approach in medicine using quantum adaptive machine learning techniques. *Optical and Quantum Electronics*, *56*(4), 534.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. (2017). Attention is all you need. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), *Advances in neural information processing systems, vol. 30*. Curran Associates, Inc., https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., & Bowman, S. R. (2018). GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP workshop blackboxNLP: Analyzing and interpreting neural networks for NLP* (pp. 353–355).

Wu, T., He, S., Liu, J., Sun, S., Liu, K., Han, Q.-L., et al. (2023). A brief overview of ChatGPT: The history, status quo and potential future development. *IEEE/CAA Journal of Automatica Sinica*, *10*(5), 1122–1136. http://dx.doi.org/10.1109/JAS.2023.123618.

Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R. R., & Le, Q. V. (2019). Xlnet: Generalized autoregressive pretraining for language understanding. *Advances in Neural Information Processing Systems*, *32*.

Yang, C.-H. H., Qi, J., Chen, S. Y.-C., Tsao, Y., & Chen, P.-Y. (2022). When BERT meets quantum temporal convolution learning for text classification in heterogeneous computing. In *ICASSP 2022 - 2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)* (pp. 8602–8606). http://dx.doi.org/10.1109/ICASSP43922.2022.9746412.

Zeng, W., & Coecke, B. (2016). Quantum algorithms for compositional natural language processing. *Electronic Proceedings in Theoretical Computer Science*, *221*, 67–75. http://dx.doi.org/10.4204/eptcs.221.8.

Zhu, Y., Kiros, R., Zemel, R., Salakhutdinov, R., Urtasun, R., Torralba, A., et al. (2015). Aligning books and movies: Towards story-like visual explanations by watching movies and reading books. In *Proceedings of the IEEE international conference on computer vision* (pp. 19–27).